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3 April 1987
SUPERSEDING
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27 March 1981

MILITARY SPECIFICATION

REGULATOR, OXYGEN, ACETYLENE, WELDING APPLICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers single and acetylene regulators and single and two stage oxygen regulators used for the controlled withdrawal of compressed oxygen and acetylene gasses from compressed gas cylinders.

1.2 Classification. Regulators shall be of the following types and sizes as specified (see 6.2.1).

Type I - Acetylene, Single-Stage
Type II - Oxygen, Single-Stage
Type III - Oxygen, Two Stage

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-C-843 - Cushioning Material, Cellulosic
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B40.1 - Gauges-Pressure, Indicating Dial Type-Elastic Element

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COMPRESSED GAS ASSOCIATION (CGA)

CGA E-1 - Standard Connections for Regulator Outlets, Torches and Fitted Hose for Welding and Cutting Equipment
CGA E-4 - Standard for Gas Regulators for Welding and Cutting
CGA V-1 - Compressed Gas Cylinder Valves Outlet and Inlet Connections
CGA G-4.1 - Cleaning Equipment for Oxygen Service

(Application for copies should be addressed to the Compressed Gas Association Inc., Crystal Gateway 1, Suite 501, 1235 Jefferson Davis Highway, Arlington, VA 22202.)

UNDERWRITER'S LABORATORIES, INC., (UL)

UL 252 - Compressed Gas Regulators
UL 404 - Gauges, Indicating Pressure, for Compressed Gas Service
UL 969 - Marking and Labeling Systems

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2.1, and 6.3).

3.2 Design. The regulator shall be new and of the manufacturer's current configuration.

3.2.1 Single stage regulators. Single stage regulators shall use a manually adjusted spring loaded diaphragm for pressure regulation.

3.2.2 Two stage regulators. Two stage regulators shall use a preset first stage with a spring loaded diaphragm or a piston and a manually adjusted second stage with a spring loaded diaphragm.

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3.3 Construction. The regulator shall be constructed of parts and materials that are without defects. Each regulator shall be delivered assembled so that it can be connected to an appropriate gas cylinder and so connected may be capable of being used for all operations for which it was designed. Regulators may be delivered with the pressure adjusting screw disassembled from the regulator.

3.3.1 Materials. Materials not specifically designated herein or in the contract shall be of quality commensurate with commercial practice within the regulator manufacturing industry, shall be suitable for the intended purpose in the design of the regulator and shall be free from defects which would adversely affect the performance of the assembly. It shall not be permissable to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.2 Forgings. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges and other surface defects that adversely affect the performance of the regulator.

3.3.3 Machine surfaces. Gas holes and passages shall be free of burrs. Machined surface seals shall be free of nicks, burrs, and scratches and shall meet the leakage requirements of 3.4.2.3.

3.3.4 Fastening devices. Fasteners shall be installed in a manner that prevents change of tightness. Those fasteners subject to removal of adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein. Inlet and outlet connection threads shall conform to CGA V-1 and CGA E-1. All internal and external threads shall be clean, free from extraneous materials, and shall have no burrs or nicks.

3.3.6 Interchangeability. All parts shall be manufactured to standards that shall permit replacement without modification of the regulator or the part being installed as the replacement.

3.3.7 Maintainability. Regulators shall be so constructed that they can be repaired by replacement of parts such as gauges, pressure adjusting devices, springs, diaphragms, connections, valve seats, etc., when replaced by qualified personnel, without causing damage to the regulator or any of its parts.

3.3.8 Cleanliness. Regulators shall be delivered cleaned for oxygen service in accordance with CGA G-4.1. This requirement applies equally to both acetylene regulators and oxygen regulators. All regulator parts in contact with the gas shall be free of oil, grease, and other foreign matter not specifically required by the manufacturer's design.

3.3.9 Safety. Regulators shall be designed and manufactured to comply with the requirements of UL 252 and CGA E-4. Thread sealing compounds and lubricants shall be compatible with the gas and pressure to which they are exposed in service.

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3.4 Performance characteristics. Regulators and all components shall perform in accordance with all the requirements of UL 252, CGA E-4, and 3.4.1 through 3.4.3 herein.

3.4.1 Flow rates (see 6.5.4). Regulators shall deliver the specified standard cubic feet per hour (scfh) as specified herein. Scfh shall be defined in 6.5.6 and correction factors shall be applied to the calculation of test results.

3.4.1.1 Single and two stage oxygen flow rate. Single and two stage oxygen regulators shall deliver oxygen at no less than 1200 scfh at 120 ±5 pounds per square inch gauge (psig) while meeting the requirements of 3.4.2.4 at no more than 300 psig inlet pressure. For testing purposes, nitrogen or oil free air shall be used with correction factors applied.

3.4.1.2 Acetylene flow rate. Acetylene regulators shall deliver acetylene at no less than 10 psig delivery pressure when the inlet pressure is no higher than 50 psig and the flow rate is no less than 275 scfh. For testing purposes, nitrogen or oil free air shall be used with correction factors applied.

3.4.2 Range of delivery pressures. Regulators shall deliver the pressure specified in Table I.

3.4.2.1 Regulator pressure regulation (see 6.5.3). For a single stage regulator, the change in a set delivery pressure shall not be more than 0.8 percent of a change in inlet pressure when the flow rate is 75 ±25 scfh for oxygen and 30 ±15 scfh for acetylene. For a two stage oxygen regulator, the change in set delivery pressure shall not be more than 0.1 percent of a change in inlet pressure when the flowrate is 75 ±25 scfh.

3.4.2.2 Chatter and vibration. The regulator shall not experience self initiated audible chatter or vibration during all normal use conditions throughout its full rated flow range.

3.4.2.3 Leakage. All regulators shall meet the leakage requirements of CGA E-4 and UL 252.

3.4.2.4 Oxygen regulator flow regulation (see 6.5.2). The change in delivery pressure resulting from an increase in flow shall not exceed 15 psig when the flow rate is changed from 75 ±25 scfh to at least 1200 scfh with the inlet pressure no more than 300 psig and the outlet pressure initially set at 120 ±5 psig.

3.4.2.5 Acetylene regulator flow regulation (see 6.5.2). The change in delivery pressure resulting from an increase in flow shall not exceed 5 psig when the flow rate is changed from 30 ±15 scfh to at least 275 scfh with the inlet pressure no more than 50 psig and the delivery pressure initially set at 15 +0-1 psig.

3.4.2.6 Static increment. The static increment (see 6.5.1) for an oxygen regulator that occurs as a result of a change in flow rate from 75 ±25 scfh to no flow shall not exceed 10 psig in 2 minutes when the delivery pressure is 100 psig and the inlet pressure is 300 psig. The static increment for an acetylene regulator that occurs as a result of a change in flow from 30 ±15 scfh to no flow shall not exceed 5 psig in 2 minutes when the delivery pressure is 10 psig and the inlet pressure is 50 psig. No further increase in static increment shall be allowed in the 5 minutes following the 2 minute period.

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3.4.3 Excess pressure. Regulators shall meet the requirements of CGA E-4 and UL 252.

TABLE I - REGULATOR PERFORMANCE AND CHARACTERISTICS

Regulator	Maximum Inlet psig	Delivery Pressure Range, psig	Flow Rate scfh	Test Pressure Inlet psig, max	Test Pressure Outlet psig, min
Type I Acetylene Single Stage	400	0-15, red lined to 30	275	50	10
Type II Oxygen Single Stage	3000	0-120 or higher	1200	300	120
Type III Oxygen Double Stage	3000	0-120 or higher	1200	300	120

3.5 Components. Regulators shall consist of the following principal components: body, bonnet(s), valve seat assembly(s), pressure responsive element(s), pressure adjusting screw(s), inlet pressure gauge, delivery pressure gauge, inlet filter, inlet connection, outlet connection and relief valve (oxygen only).

3.5.1 Body. Regulator bodies shall be fabricated of brass. In no instance shall the copper content of an acetylene regulator body exceed 70 percent. All bodies shall be machined from a single piece forging or bar stock.

3.5.2 Bonnet. The bonnet for single stage regulators, and the first and second stage bonnets for two stage regulators shall be fabricated of metal. The bonnet of the single stage regulator and the second stage bonnet of the two stage regulator shall be provided with threads to retain and position the pressure adjusting screw. Non-metallic thread inserts are acceptable. Bonnets shall be vented.

3.5.3 Valve seat assembly. The valve seating components shall be so supported as to maintain alignment under all performance conditions specified herein. The valve seat material shall be compatible with the gas and pressure to which they will be exposed in service.

3.5.4 Pressure responsive element. The pressure responsive element shall be fabricated of corrosion resistant metal or an elastomer compatible with the gas and pressure to which it will be exposed in service. The pressure responsive element shall be replaceable without the application of heat.

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3.5.5 Pressure adjusting screw. The travel of the pressure adjusting screw for the acetylene regulator shall be restricted to permit a maximum pressure setting, at no flow condition, of not more than 30 psig. Adjusting screws shall be fabricated of material that is corrosion resistant or plated to resist corrosion. The pressure adjusting screw shall be hand operable.

3.5.6 Pressure indicating gauges. Pressure indicating gauges shall be of the elastic element type in accordance with ANSI B40.1 including Appendix B and UL 404 where applicable, with graduations as specified herein. Unless otherwise specified (see 6.2.1) all indicating gauges shall be marked in the US Customary System of Units (in this case "psi") and also in the Standard International System of Units (SI) (in this case kilopascals "kPa") in accordance with FED-STD-376. All gauges shall be installed to meet the leakage requirements specified herein (see 3.4.2.3). All gauges shall be solid or open front with a pressure relief opening and solid brass cases with lenses retained by a screw on bezel or threaded plastic lenses that screw onto the case. Gauges shall be stem mounted to the regulator body with American Standard taper pipe threads (NPT). Gauges shall be for outdoor field use, resistant to 99 percent humidity, weather and dust. All gauges shall be 2 inches nominal size or larger.

3.6 Filter. A 60 micron or finer mesh filter shall be located in the inlet passage before the valve seat. The filter material shall be corrosion resistant, compatible with the gas and pressure to which it will be exposed in service, and replaceable.

3.7 Inlet and outlet connections.

3.7.1 Oxygen inlet. Unless otherwise specified (see 6.2.1), oxygen inlet connections shall be in accordance with CGA V-1 connection no. 540.

3.7.2 Oxygen outlet. Unless otherwise specified (see 6.2.1), oxygen outlet connections shall be in accordance with CGA E-1, connection no. 022.

3.7.3 Acetylene inlet. Unless otherwise specified (see 6.2.1), acetylene inlet connections shall be in accordance with CGA V-1 connection no. 510.

3.7.4 Acetylene outlet/delivery. Unless otherwise specified (see 6.2.1), acetylene outlet/delivery connections shall be in accordance with CGA E-1, connection no. 023.

3.7.5 Alternate acetylene cylinder connection adapter. When specified (see 6.2.1), a brass adapter shall be furnished with the regulator (unattached) to convert a CGA 510 inlet connection to a CGA 300 inlet valve connection (see CGA V-1).

3.7.6 Relief valve. At least one self resetting relief valve shall be installed on single and two stage oxygen regulators. The valve(s) shall relieve 200 ±50 psig pressure from the delivery pressure side of the regulator. However, if a valve is installed in the first stage of a two stage regulator, the valve shall be set to relieve pressure according to the manufacturer's standard practice to protect the regulator from over pressure and no valve is then required to be installed in the second stage. Relief valves are not required in acetylene regulators.

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3.8 Product identification. Regulators and gauges shall be marked in accordance with UL 969 and MIL-STD-130, and, unless otherwise specified regulator marking shall include the National Stock Number. Markings shall include the regulator manufacturer's name and model number.

3.9 Finish. The finish on the regulator shall be equivalent to the standard commercial practice within the regulator manufacturing industry.

3.10 Workmanship. Workmanship of the regulators shall be of the highest quality prevailing among manufacturers normally producing regulators of the type specified herein for the commercial market.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified (see 6.2.1), first article inspection shall be performed on the preproduction model(s) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.5, and the tests of 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (g) as follows, and failure of any unit to pass an examination or test shall be cause for rejection of the unit.

a. Product examination (see 4.5).

b. Pressure gauge accuracy test (see 4.6.1).

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c. Seat leak test (see 4.6.2).

d. Overall leak test (see 4.6.3).

e. Flow capacity and regulation test (see 4.6.4).

f. Static increment test (see 4.6.5).

g. Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105, using the following:

Inspection or test	Level	AQL
Product examination	II	1.0
Pressure gauge accuracy test	II	1.0
Seat leak test	II	1.0
Overall leak test	II	1.0
Flow capacity and regulation test	II	1.0
Static increment test	II	1.0
Packaging inspection	S-4	4.0

4.5 Product examination. Visually, dimensionally, and manually examine each regulator to determine conformance with the requirements of 3.2 through 3.3.9, 3.5 through 3.7.5, and 3.8 through 3.10. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Dimensional examination includes measuring dimensions including weights, as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the regulator and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.3.6), shall be cause for rejection.

4.6 Tests. Tests are to be conducted using pressure gauges calibrated as stated in paragraph 4.6.1, and flow meters having an accuracy of not less than 3 percent of measured value. A control valve shall be provided on each side of the regulator valve. Oil-free air or nitrogen shall be used for all tests. Flows listed are for air. Standard cubic feet per hour (scfh) shall be as defined in 6.5.6 and correction factors shall be applied as required. Failure of the regulator to pass any of the following tests shall be cause for rejection.

4.6.1 Pressure gauge accuracy calibration. Pressure gauge accuracy shall be tested by calibrating as follows: using a standard pressure measuring device of applicable type and range, and having an overall accuracy of 0.25 percent, apply pressure at one major graduation in each 1/4 portion of the scale. The accuracy of all gauges shall be within 3 percent in the lower and upper 1/4 portions of the scale and within 2 percent in the center 1/2 portion of the scale (see 3.5.6).

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4.6.2 Seat leak. With the adjusting screw completely released, apply 2000 psig minimum pressure to the inlet of the oxygen regulator or 250 psig minimum pressure to the inlet of the acetylene regulator. There shall be no leakage past the high pressure seat when observed for a period of 5 minutes while using a suitable leak detection method (see 3.4.2.3).

4.6.3 Overall leak. With the regulator attached to the proper mating fittings, apply 2000 psig minimum pressure to the oxygen inlet or 250 psig minimum pressure to the acetylene inlet. With the outlet blocked, adjust the oxygen outlet pressure to 120±5 psig or the acetylene outlet pressure to 14±1 psig. There shall be no leakage from the regulator when observed for a period of 5 minutes while using a suitable leak detection method (see 3.4.2.3). This includes the threaded connections, diaphragm clamping surfaces, pressure relief device and pressure gauges.

4.6.4 Flow capacity and regulation. Set inlet pressure, outlet pressure, and flow per table II. Increase flow to minimum specified. Regulator shall allow minimum flow specified with a maximum change in pressure as specified (see 3.4.2.4 and 3.4.2.5).

TABLE II - Flow Capacity and Regulation

Gas Circuit Tested	Set inlet Pressure to max of psig	Set Pressure to: psig	Set Flow to: scfh	Increase Flow to: Minimum scfh	Maximum allowed Pressure Change psig
Oxy Acetylene	300 50 1/	125±5 14 to 15	75±25 30±15	1200 275	15 5

1/ Cylinder may be manifolded.

4.6.5 Static increment. Set inlet pressure flow per table III. Shut off flow and observe for pressure increase as listed (see 3.4.2.6).

TABLE III - Static Increment

Gas Circuit Tested	Set Inlet Pressure to min of: pisg	Set Flow to: scfh	Change Flow to: scfh	Maximum Pressure Increase, psig, in:	
				First 2 min	Next 5 min
Oxy Acetylene	300 50	75±25 30±15	0 0	10 5	0 0

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4.6.6 Pressure regulation. Set inlet pressure flow as listed in table IV. Reduce inlet pressure and determine percentage change in outlet pressure (see 3.4.2.1).

TABLE IV - Pressure Regulation.

Type Reg	Set Inlet Pressure to min of: psig	Set Flow to: scfh	Change Inlet Pressure to max of: psig	Maximum Allowed Change in Outlet Pressure psig
Single Stage				
Oxy	2500	75±25	500	16.0
Acet	350	30±15	50	2.4
Two Stage				
Oxy	2500	75±25	500	2.0

4.6.7 Excess pressure. The regulators shall be tested for excess pressure in accordance with the excess pressure test in UL 252. The regulators shall also meet all the requirements of UL 252 (see 3.4.3).

4.7 Packaging inspection. The regulator shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirements specified in Section 5.

5 PACKAGING

5.1 Preservation. Preservation shall be level A or industrial as specified in the contract (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The regulators shall be cleaned and dried in accordance with MIL-P-116. Cleaning shall be in accordance with process C-1 and drying shall be in accordance with any applicable procedure of MIL-P-116.

5.1.1.2 Unit protection. The regulators shall be individually processed in accordance with method III of MIL-P-116. Openings shall be sealed with tape conforming to PPP-T-60, type II or caps and plugs conforming to MIL-C-5501. The gauges and any projecting parts of the regulator shall be wrapped with cushioning material conforming to PPP-C-843, type II, in a manner to prevent damage of the gauges or regulators and to restrict movement within the container. The cushioned item shall be placed in a fiberboard container conforming to PPP-B-636, class weather-resistant. Technical data issued with the regulators shall be processed method 1C-1 of MIL-P-116, and placed inside the fiberboard box. Closure shall be in accordance with the appendix of the container specification.

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5.1.2 Industrial. When industrial packaging is specified, preservation shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, level B, or industrial as specified in the contract (see 6.2.1).

5.2.1 Level A. Regulators and publications provided unit protection as specified in 5.1.1.2 shall be packed in quantities utilized when the gross weight of the shipping container does not exceed the size and weight limitations provided for in the applicable container specification. The shipping container shall conform to the requirements of PPP-B-601, overseas type, or PPP-B-621, class 2. Arrangement of the contents, cushioning, blocking, bracing and anchoring shall be in accordance with MIL-STD-1186. Closure and strapping of the shipping container shall be in accordance with the appendix of the applicable shipping container specification.

5.2.2 Level B. Regulators and publications provided unit protection as specified in 5.1.1.2 shall be packed in quantities utilized when the gross weight of the shipping container does not exceed the size and weight limitations provided for in the applicable container specification. The shipping container shall conform to the requirements of PPP-B-640, class 2, style E. Arrangement of the contents, cushioning, blocking, bracing and anchoring shall be in accordance with MIL-STD-1186. Closure and strapping of the shipping container shall be in accordance with the appendix of the shipping container specification.

5.2.3 Industrial. Regulators preserved as specified in 5.1.2 shall be packed in accordance with ASTM D3951.

5.3 Marking. Marking shall be military or industrial, as specified (see 6.2.1).

5.3.1 Military. In addition to any special markings specified by the contract or purchase order, unit containers and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Industrial. When industrial packaging is specified, marking of unit containers and exterior shipping containers shall be marked in accordance with ASTM D3951.

6 NOTES

6.1 Intended use. The regulators covered by this specification are intended for use with oxy-acetylene welding and cutting torches, metallizing torches, laboratory equipment, etc. The regulators are for use with all types of torches. The single stage regulators are used where an occasional adjustment of the delivery pressure is required to maintain a constant delivery pressure, this adjustment not being detrimental to the operation. The two-stage regulators are used when a unattended constant delivery pressure is required with decreasing inlet pressures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date for this specification.

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b. First article, when required (see 3.1).

c. Calibrations (SI or dual US and SI), if required (see 3.5.6).

d. Oxygen inlet, if different (see 3.7.1).

e. Oxygen outlet, if different (see 3.7.2).

f. Acetylene inlet, if different (see 3.7.3).

g. Acetylene outlet, if different (see 3.7.4).

h. Brass adapter, when required (see 3.7.5).

i. First article inspection, if different (see 4.3).

j. Levels of preservation, and packing required (see 5.1 and 5.2).

k. Type of marking required (see 5.3).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated in the contract.

6.3 First article. When a first article inspection is required, the regulator(s) should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering products which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government is presently appropriate for the pending contract.

6.4 Repair part manuals. Repair parts identification, repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Commercial and Government Entity (CAGE), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.5 Definitions.

6.5.1 Static increment. The increase in pressure resulting from a reduction in flow from a relatively small flow to no flow.

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6.5.2 Flow regulation. A change in delivery pressure resulting from a change in flow.

6.5.3 Pressure regulation. A change in delivery pressure resulting from a change in inlet pressure.

6.5.4 Flow capacity. Flow capacity is the maximum flow rate the regulator will deliver at a specified inlet pressure within a specified change in delivery pressure.

6.5.5 Maximum delivery pressure. Maximum delivery pressure is the delivery pressure guaranteed by the manufacturer under static conditions. The regulator will deliver at least the maximum delivery pressure and may also deliver more.

6.5.6 Standard cubic feet. Standard air is defined in the Compressed Air and Gas Handbook, copyright 1973, by the compressed Air and Gas Institute, as air at a temperature of 68 degrees Fahrenheit (F), a pressure of 14.70 pounds per square inch absolute (psia) and a relative humidity of 36 percent and a density of .0750. Standard cubic feet per hour is the delivery rate of standard air.

6.6 Subject term (key word) listing.

Acetylene
CGA fittings
Compressed gas
Flow, Rate
Flow, Regulator
Oxygen
Pressure, Regulation
Regulator
Static increment
Welding

6.7 Changes from previous issue. Asterisks are not used in the revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Navy - YD
Air Force - 99

Review activities:

Air Force - 84

Preparing activity

Army - AL

Project No. 4820-0520

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-13877 D	REGULATOR, OXYGEN, ACETYLENE, WELDING APPLICATION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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